ITTNA3-22 4IR Assignment Writing Template

Distinction-focused workbook for planning, drafting, citations, and rubric checks

Academic safety note: do not submit this workbook as your assignment. Use it to plan, write, and check your own answer in your own wording. The brief explicitly warns against submitting AI-generated content as your own work.

Assignment Snapshot

- Module: ITTNA3-22, 4IR Technologies
- Assessment: Individual assignment
- Total: 100 marks
- Suggested length: 2,800-3,200 words unless your lecturer gives a different limit
- Referencing: Harvard style, used consistently
- Main argument to develop: 4IR technologies can improve education, research, employment, productivity, and household income in Africa and South Africa, but the benefits depend on solving inequality, infrastructure, skills, and inclusion gaps.

Suggested Word Budget

Section	Marks	Suggested words	Purpose
Introduction	0	150-200	Frame 4IR and state your overall argument
Q1.1	20	600-700	Discuss four inequalities, impacts, examples, strategies
Q1.2	20	550-650	Present digital-tech project and mentorship model
Q1.3	30	850-950	Compare Japan and South Africa, then justify robotics benefits
Q1.4	30	750-850	Explain sharing economy and disposable income
Conclusion	0	100-150	Tie the assignment together

Source Bank

Use the assignment scenario as your main source, then add supporting sources where factual claims need backing.

Use for	Source	Notes to cite
Assignment context	Via Afrika podcast and scenario in assignment PDF	4IR impact on education, African universities, Japan robotics, sharing economy
South African unemployment	Statistics South Africa, QLFS Q1 2026	National unemployment was 32.7%; youth unemployment remained especially severe
Internet access inequality	Statistics South Africa, General Household Survey 2025	Fixed home internet access remains much lower than general/mobile access
African higher education	UNESCO higher education in Africa article	R&D underinvestment, gender disparities in STEM, need for modern curricula and industry partnerships
Robotics in Japan	International Federation of Robotics article	Japan's 2024 logistics/labour problem and robot support in dirty, dull, dangerous work
South African skills gap	WEF Future of Jobs Report 2025	Skills gaps and emerging roles such as AI, machine learning, robotics, and digital transformation
Sharing economy	Harvard Kennedy School sharing-economy article	Internet-based markets allow owners to rent durable goods to non-owners

Introduction Writing Plan

Write one short paragraph.

Include:

- A plain definition of 4IR.
- A sentence linking 4IR to education, work, business, society, and households.
- Your overall argument.

Build your paragraph from these points:

- 4IR combines technologies such as AI, robotics, IoT, cloud computing, biotechnology, and digital platforms.

- These technologies change how people learn, research, work, trade, and access services.
- In Africa and South Africa, 4IR is not automatically inclusive.
- The assignment will argue that 4IR can create development opportunities only when inequality, skills gaps, infrastructure, and exclusion are addressed.

Your own paragraph:

[Write 5-7 sentences here.]

Q1.1: Four Inequalities In African Universities

Question focus: Discuss the impact of any four inequalities on innovation and research in African institutions and suggest strategies to overcome them.

High-mark structure: use four mini-sections. For each inequality, cover:

1. What the inequality is.
2. How it weakens innovation and research.
3. African or South African example.
4. Practical strategy.

Inequality 1: Digital Access Inequality

Key points to transform into your own paragraph:

- Not all students and researchers have equal access to devices, reliable internet, affordable data, stable electricity, and learning platforms.
- This limits online learning, access to digital libraries, research databases, cloud tools, virtual labs, and collaboration with other institutions.
- In South Africa, many households access the internet through mobile connections, while fixed home internet remains much less available. This matters because research work often needs stable, high-volume connectivity rather than only mobile data.
- Practical strategies include device-loan schemes, zero-rated learning platforms, campus Wi-Fi zones, negotiated student data bundles, open computer labs, and community digital access centres.

Citation slot:

(Statistics South Africa, 2025)

Your own paragraph:

[Write 7-9 sentences here. Start with the inequality, then explain impact, example, and strategy.]

Inequality 2: Research Funding And Infrastructure Inequality

Key points to transform into your own paragraph:

- Some African universities lack advanced laboratories, modern equipment, licensed software, cloud credits, reliable research funding, and technology transfer support.
- This can prevent researchers from testing ideas, building prototypes, collecting quality data, publishing competitively, or commercialising inventions.
- UNESCO notes that underinvestment in African higher education research and development limits innovation capacity, especially in fields such as health, agriculture, and technology.
- Practical strategies include shared regional research hubs, government grants, industry-funded labs, open innovation challenges, cloud-credit partnerships, and stronger university-business collaboration.

Citation slot:

(UNESCO, 2024)

Your own paragraph:

[Write 7-9 sentences here.]

Inequality 3: Skills And Curriculum Inequality

Key points to transform into your own paragraph:

- Some institutions update curricula faster than others, meaning students may not receive enough exposure to AI, robotics, data analytics, IoT, cybersecurity, cloud platforms, and ethical technology use.
- This weakens innovation because researchers and graduates may understand theory but lack practical tools for 4IR problem-solving.
- South African employers identify skills gaps as a major barrier to business transformation, which shows why universities need to connect learning with industry needs.
- Practical strategies include lecturer upskilling, industry co-designed modules, short professional certificates, hackathons, work-integrated learning, and project-based assessments.

Citation slot:

(World Economic Forum, 2025)

Your own paragraph:

[Write 7-9 sentences here.]

Inequality 4: Gender And Socio-Economic Inequality

Key points to transform into your own paragraph:

- Women, poorer students, and historically disadvantaged communities may have less access to STEM subjects, technology clubs, funding, role models, safe learning spaces, and professional networks.
- This affects research and innovation because diverse teams are more likely to understand local problems and design inclusive solutions.
- If disadvantaged groups are excluded, universities lose potential innovators who could solve problems in communities similar to their own.
- Practical strategies include targeted scholarships, women-in-STEM programmes, mentorship, inclusive innovation competitions, childcare support where relevant, and outreach into under-resourced schools.

Citation slot:

(UNESCO, 2024)

Your own paragraph:

[Write 7-9 sentences here.]

Q1.1 Mini-Conclusion

Use 2-3 sentences to connect the four inequalities.

Point to make:

- These inequalities are connected: poor connectivity limits skills development; weak funding limits infrastructure; exclusion reduces the diversity of ideas.
- African universities need combined strategies, not isolated fixes.

Your own mini-conclusion:

[Write 2-3 sentences here.]

Q1.1 checklist:

- Four inequalities are clearly named.
- Each inequality explains impact on innovation and research.
- Each inequality has an African or South African example.
- Each inequality has a practical strategy.
- The answer is structured and logical.

Q1.2: Digital Technology Project

Question focus: Assume you are one of the 21 African leaders in AfricaInLead. Choose bio or digital technology and outline a project idea that addresses a socio-economic or environmental challenge in your community. Explain mentorship for sustainability and growth.

Chosen technology: Digital technology.

Project idea: Urban Digital Skills and Micro-Innovation Hub.

Project Context

Key points to transform into your own paragraph:

- The local issue is urban youth unemployment, digital exclusion, lack of work experience, and weak access to professional networks.
- Many young people may own or access a phone but still lack deeper digital skills required for paid work.
- The project addresses the gap between basic digital access and employable digital capability.
- Link this to Stats SA unemployment data and WEF skills-gap findings.

Citation slots:

(Statistics South Africa, 2026)
(World Economic Forum, 2025)

Your own paragraph:

[Write 6-8 sentences here.]

How The Project Works

Key points to transform into your own paragraph:

- The hub offers practical short courses in coding basics, digital marketing, data literacy, cybersecurity awareness, cloud collaboration, and entrepreneurship.
- Learners work in teams on real community projects.
- Example projects: a website for a township business, a booking form for a local service provider, a simple stock tracker for a spaza shop, or a dashboard for a community organisation.
- AI tools can support learning through practice questions, language support, and coding help, but mentors teach responsible and ethical use.
- The hub should be mobile-first because smartphones are more common than fixed home internet.

Your own paragraph:

[Write 8-10 sentences here.]

Community Impact

Key points to transform into your own paragraph:

- Youth gain practical skills and a portfolio of completed work.
- Local businesses get affordable digital support.
- The community benefits from more digitally capable young people and small businesses.
- The project supports entrepreneurship because learners can offer basic services after training.
- Impact should be measured using completions, portfolios, internships, client projects, small ventures, and job placements.

Your own paragraph:

[Write 7-9 sentences here.]

Mentorship And Sustainability

Key points to transform into your own paragraph:

- Use a train-the-trainer model: top graduates return as peer mentors.
- Partner with universities, local businesses, NGOs, and technology professionals.
- Run monthly showcase days where learners present projects to potential employers or clients.
- Create mentorship circles: technical mentor, career mentor, and community project mentor.

- Keep the project sustainable through sponsorships, small business service fees, alumni volunteering, and local government or university support.

Your own paragraph:

[Write 7-9 sentences here.]

Q1.2 checklist:

- Digital technology is clearly identified.
- The socio-economic problem is specific to an urban South African context.
- The project is realistic and creative.
- The answer explains how the project works.
- Community impact is clear.
- Mentorship is practical and sustainable.

Q1.3: Japan And South Africa Robotics Comparison

Question focus: Compare Japan and South Africa, then explain how robots can still benefit South Africa despite high unemployment.

High-mark structure: make the comparison explicit. Do not write only about Japan first and South Africa later without linking them.

Opening Comparison

Key points to transform into your own paragraph:

- Japan and South Africa both face labour-market pressure, but the nature of the pressure is different.
- Japan faces labour shortages linked to an ageing population, logistics pressure, and working-time regulations.
- South Africa faces high unemployment, especially youth unemployment, plus skills mismatch and inequality.
- This means robotics has a different purpose in each country: in Japan it fills gaps where workers are scarce; in South Africa it must improve productivity while creating pathways into new forms of work.

Citation slots:

(International Federation of Robotics, 2023)
(Statistics South Africa, 2026)

Your own paragraph:

[Write 7-9 sentences here.]

Japan: Why Robots Make Sense There

Key points to transform into your own paragraph:

- Japan's 2024 labour problem involved logistics and working-time limits for truck drivers.
- Robots can support loading, unloading, warehouse movement, manufacturing, and service tasks.
- IFR argues that robots can reduce dirty, dull, and dangerous work and help close productivity gaps.
- In Japan, robots are easier to justify because the problem is often too few available workers rather than too many unemployed people.

Citation slot:

(International Federation of Robotics, 2023)

Your own paragraph:

[Write 7-9 sentences here.]

South Africa: Why The Issue Is More Complex

Key points to transform into your own paragraph:

- South Africa's unemployment rate is high, and youth are heavily affected.
- Introducing robots without planning could worsen job anxiety or displace low-skilled workers.
- However, the country also has productivity, safety, infrastructure, and service-delivery challenges.
- The key argument is that South Africa should use robotics for augmentation and new value creation, not simple labour replacement.

Citation slot:

(Statistics South Africa, 2026)

Your own paragraph:

[Write 7-9 sentences here.]

Similarities And Differences

Use this comparison table to build one analytical paragraph.

Point	Japan	South Africa
Main labour problem	Labour shortage and ageing workforce	High unemployment and skills mismatch
Main robotics purpose	Fill labour gaps and protect overworked workers	Improve productivity, safety, and create skilled technical roles
Main risk	Dependence on automation and cost	Job displacement if workers are not reskilled
Best policy response	Automation in shortage sectors	Automation plus training, local manufacturing, and job transition support

Your own paragraph:

[Write 7-9 sentences comparing the countries directly.]

Benefits For South Africa Despite Unemployment

Choose at least four examples and explain them properly.

Example bank:

- **Mining:** robots and remote-controlled systems can reduce exposure to dangerous underground conditions.
- **Logistics and warehousing:** robots can improve stock movement, reduce delays, and create technician and maintenance roles.
- **Agriculture:** drones, sensors, and automated systems can improve irrigation, crop monitoring, and food production.
- **Healthcare:** robots can support repetitive delivery tasks in hospitals, allowing nurses to focus on patient care.
- **Manufacturing:** robotics can improve product quality, reduce waste, and help local firms compete.
- **Education and training:** robotics clubs, TVET programmes, and university labs can prepare students for technical jobs.

Key point:

- The strongest answer does not pretend robotics has no risk. It argues that robotics is beneficial when linked to reskilling, worker participation, local support services, and inclusive policy.

Citation slot:

(World Economic Forum, 2025)

Your own paragraph:

[Write 10-12 sentences here.]

Q1.3 Closing Judgement

Point to make:

- Japan shows how robots can solve labour shortages, but South Africa must adapt robotics to a different labour reality.
- Robotics can benefit South Africa if it is treated as part of industrial upgrading and skills development rather than as a shortcut to reduce labour costs.

Your own paragraph:

[Write 3-4 sentences here.]

Q1.3 checklist:

- Japan and South Africa are directly compared.
- Japan's labour shortage is explained.
- South Africa's unemployment context is explained.
- The answer includes similarities and differences.
- Benefits for South Africa are justified with examples.
- Risks and mitigations are included.

Q1.4: Sharing Economy And Disposable Income

Question focus: Examine how sharing underutilised assets can increase disposable income for households in the sharing economy.

Define The Sharing Economy

Key points to transform into your own paragraph:

- The sharing economy uses digital platforms to connect people who own underused assets with people who need temporary access.
- Assets can include spare rooms, cars, tools, bicycles, books, Wi-Fi, solar power, or skills.
- Ownership does not always change; instead, people rent, lend, share, or exchange temporary access.
- Link this to the assignment scenario's examples of Airbnb, cars, tools, books, bicycles, peer-to-peer lending, and skills.

Citation slots:

(Van Embden, 2018)
(Filippas, Horton and Zeckhauser, 2020)

Your own paragraph:

[Write 6-8 sentences here.]

How Asset Owners Increase Disposable Income

Key points to transform into your own paragraph:

- Disposable income is money left for households after essential expenses or after earning extra income.
- Underutilised assets often sit idle even though they still cost money to own or maintain.
- Renting out an asset converts idle capacity into income.
- Example: a spare room can generate short-term rental income; a car can earn income through e-hailing or car sharing; power tools can be rented to people doing repairs.
- This extra income can help pay for food, electricity, transport, school fees, debt, or savings.

Your own paragraph:

[Write 8-10 sentences here.]

How Users Save Money

Key points to transform into your own paragraph:

- The sharing economy also increases disposable income by reducing the need to buy expensive items.
- A household may not need to buy a drill, bicycle, car, textbook, or generator if it can access one only when needed.
- Lower upfront costs make it easier for people to start small income-generating activities.
- Example: someone doing a handyman job can rent tools for a day instead of buying a full toolkit.
- This lowers barriers to entry for informal work and small businesses.

Your own paragraph:

[Write 7-9 sentences here.]

Community And Business Examples

Use 3-5 examples and explain each in relation to disposable income.

Example bank:

- **Homes:** spare rooms rented to students, tourists, or workers.
- **Cars:** e-hailing, delivery work, or community car-sharing.
- **Tools:** tool libraries or rentals for repairs and informal jobs.
- **Bicycles:** low-cost local transport for learners, workers, or delivery services.
- **Skills:** tutoring, coding, design, repairs, farming knowledge, or language services shared through platforms.
- **Energy or Wi-Fi:** community-based access to solar charging or internet hotspots.

Your own paragraph:

[Write 8-10 sentences here.]

Critical Balance

Key points to transform into your own paragraph:

- The sharing economy does not benefit everyone equally.
- People who own valuable assets may benefit more than people with no assets to share.
- Risks include platform fees, safety, damage, insurance, tax obligations, unfair ratings, and digital exclusion.
- For South African households, trust, affordable internet, consumer protection, and fair regulation are important.
- A strong answer recognises both benefits and limitations.

Your own paragraph:

[Write 7-9 sentences here.]

Q1.4 Closing Judgement

Point to make:

- Sharing underutilised assets can increase disposable income by creating extra earnings and lowering purchase costs.
- The model works best when access, trust, digital inclusion, and fair rules are in place.

Your own paragraph:

[Write 3-4 sentences here.]

Q1.4 checklist:

- Sharing economy is clearly defined.
- Underutilised assets are named.
- Disposable income is explained directly.
- Examples include cars, homes, tools, or similar assets.
- Benefits and limitations are both discussed.

Conclusion Writing Plan

Write one short paragraph.

Include:

- 4IR creates opportunities, but the benefits are uneven.
- African universities need inclusive access, funding, skills, and participation.
- Digital projects can address youth unemployment when linked to mentorship.
- Robotics and sharing platforms can support productivity and income if implemented responsibly.

Your own conclusion:

[Write 5-6 sentences here.]

Harvard-Style Reference List Starter

Check your lecturer's exact Harvard requirements and adjust punctuation if needed.

Filippas, A., Horton, J.J. and Zeckhauser, R.J. 2020. Owning, Using, and Renting: Some Simple Economics of the Sharing Economy. Management Science, 66(9), pp.4152-4172. Available at: https://www.hks.harvard.edu/publications/owning-using-and-renting-some-simple-economics-sharing-economy-0 [Accessed 4 June 2026].

International Federation of Robotics. 2023. Robots help to solve Japan's 2024 problem. Available at: https://ifr.org/news/robots-help-to-solve-japans-2024-problem/ [Accessed 4 June 2026].

Statistics South Africa. 2025. General Household Survey, 2025. Available at: https://www.statssa.gov.za/publications/P0318/P03182025.pdf [Accessed 4 June 2026].

Statistics South Africa. 2026. South Africa's youth and the labour market in Q1 2026. Available at: https://www.statssa.gov.za/?p=19526 [Accessed 4 June 2026].

UNESCO. 2024. What you need to know about higher education in Africa. Available at: https://www.unesco.org/en/articles/what-you-need-know-about-higher-education-africa [Accessed 4 June 2026].

Van Embden, E. 2018. Growing the sharing economy. Available at: https://www.bizcommunity.com/Article/196/706/183302.html [Accessed 4 June 2026].

Via Afrika. 2020. Getting to teaching in the 4IR. Available at: https://creators.spotify.com/pod/show/viaafrika/episodes/Getting-to-teaching-in-the-4IR-eb6kcm [Accessed 4 June 2026].

World Economic Forum. 2025. The Future of Jobs Report 2025. Available at: https://www.weforum.org/publications/the-future-of-jobs-report-2025/ [Accessed 4 June 2026].

Final Distinction Checklist

Before submission, check:

- The file name follows the brief: ITTNA3-22 - Individual Assignment - Campus Name - Student Number.
- Each question is clearly labelled as 1.1, 1.2, 1.3, and 1.4.
- Q1.1 has four inequalities, not three and not five.
- Q1.2 clearly chooses digital technology.
- Q1.3 compares Japan and South Africa throughout.
- Q1.4 directly explains disposable income, not only sharing platforms.
- Every statistic or source-based claim has an in-text citation.

- Every in-text citation appears in the reference list.
- The assignment is written in your own consistent voice.
- You have proofread for grammar, paragraph flow, and repetition.
- You have not copied full sentences from this workbook or from sources without quotation and citation.